

MGT 452: NEW PRODUCT DEVELOPMENT

Full Time & Flex Evening MBA
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Michael W. Meyer
Associate Teaching Professor of Design

office: 3W120
email: mwmeyer@ucsd.edu
mobile: (781) 248-7550

teaching
assistant: TBA
email: TBA

Thursdays 2:00pm – 4:50pm 3E107

COURSE DESCRIPTION

The New Product Development (NPD) course is based upon the instructor's experience as a practitioner, and with the view that the student's role after graduation will likely be that of manager / participant / facilitator of NPD efforts. To that end, we will work to give the student a broad framework of the process, as well as specific tools and techniques that can be used along the way.

We will take a user-centered approach to NPD, drawing heavily on the idea of empathic design to gain a visceral understanding of what people want in a product. The emphasis will be on qualitative (vice quantitative) research techniques aimed at uncovering the "big picture" of what new product you should be developing, as well as for detailing form, features, and functionality of a specific product under development.

Our goal is to be able to define the ecosystem in which product and user live, so we can craft a complete solution that is more valuable to consumers and more difficult for competitors to dislodge, much like Apple's iPod / iTunes versus standalone MP3 players. This approach should also make our new product vision more explicit and more easily communicated to others in the organization, to gain resources and rally support.

Course materials are a mix of case studies, non-business-text readings (such as the works of Edward Tufte), multimedia (such as the films of Ray and Charles Eames), and practical exercises. Active participation is vital, as much of this can only be learned by doing.

There is a final project, in which student teams will craft a product and the accompanying explanation of why this is a compelling product for both user and business. We will use information from the practical exercises as partial input to the process, so all teams will work in the same general category, though outside supplemental work is expected.

COURSE OBJECTIVES

1. Understand the overall timeline / framework for NPD.
2. Understand and be able to apply user research techniques to inform the development process. Be able to abstract the data into useful insights, to uncover feature requirements and define the ecosystem in which the product will live.
3. Understand and be able to apply generative techniques to define and refine the core product and extended product.
4. Understand and be able to apply prototyping and visualization techniques to communicate the product vision.
5. Understand organizational issues in NPD

COURSE MATERIALS

The course will draw upon **Ulrich and Eppinger's textbook** to provide an understanding of the basics of new product development, and upon Don Norman's foundational book **The Design of Everyday Things** for a solid grounding in Cognitive Design. Cases, articles, and extracts are provided in the course reader as the basis of class discussion. We will also draw from other **materials listed under the "Required" list below**. Additional materials will be distributed as required for practical learning exercises.

Required

- Product Design and Development, 7th Edition, Karl Ulrich & Steven Eppinger, McGraw Hill, 2020
- The Design of Everyday Things, Revised and Expanded Edition, Don Norman, Basic Books, 2013
- The Market Research Toolbox: A Concise Guide for Beginners, 4th Edition, Edward F. McQuarrie, Sage Publications
- Textbooklet: Visual and Statistical Thinking: Displays of Evidence for Making Decisions, Edward Tufte, Graphics Press
- Essay: The Cognitive Style of Powerpoint, Edward Tufte, Graphics Press

Note: The Market Research Toolbox was substantially rewritten in the fourth edition. If you try to use an earlier edition, you will miss material, and the chapter assignments will not match up.

Course Reader

- Henry Heinz: Making Markets for Processed Foods, HBS 9-801-289
- "Spark Innovation Empathic Design," Dorothy Leonard and Jeffrey Rayport, Harvard Business Review, Nov-Dec 1997, #97606
- "Naked Truth Meets Market Research: Perfecting a New Shower Head? Try Watching People Shower." Washington Post Article, Sunday, February 24, 2002.
- "Margaret Mead Meets Consumer Fieldwork," Jennifer McFarland, Harvard Management Update, August 2001, Product U0108C
- Sony AIBO: The World's First Entertainment Robot, HBS 9-502-010
- Innovation at 3M Corp. (A), HBS 9-699-012
- Note on Lead User Research, HBS 9-699-014
- "Customers as Innovators: A New Way to Create Value," Stefan Tomke & Eric von Hippel, Harvard Business Review, April 2002, Product R0204F
- "Enlightened Experimentation: The New Imperative for Innovation," Stefan Tomke, Harvard Business Review, Feb 2001, Product R0102D
- "R&D Comes to Services: Bank of America's Pathbreaking Experiments," Stefan Tomke, Harvard Business Review, April 2003, Product R0304E

Resources

If you are thinking of building a basic bookshelf to bolster your NPD skills, here is my recommendation. It covers a wide range of topics, reflecting the cross-functional nature of NPD, and focuses on the aspects that will give you a competitive advantage, as opposed to the aspects that are well-understood. The links to amazon.com are for information only.

[The Evolution of Useful Things](#)

Henry Petroski

How some common items got to be the way they are - the changes they went through, and why these changes happened.

[Middletown: A Study in American Culture](#)

Robert and Helen Lynd

A rich study of the state and dynamics of turn-of-the-century America - a reference piece. Instructive in its content, as well as its blending of objective, subjective, quantitative, and qualitative information to give a complete picture.

[Guns, Germs, and Steel](#)

Jared Diamond

A reference example of thinking and explaining in an insightful and analytical manner. Takes a complex issue of human societal development, and goes beyond the usual superficial analysis to find root cause and drivers.

[Ethnography](#)

Martyn Hammersly and Paul Atkinson

A primer on what it is and how to do it.

[Customer Visits](#)

Edward McQuarrie

In the same vein as Toolbox (written for non-marketing folks - R&D managers, engineering or ops managers, etc), on a single tool. Much of what he says about how to speak with customers to elicit useful, less-biased information is broadly applicable to other methods of inquiry.

[How to Design Surveys](#)

[How to Ask Survey Questions](#)

Arlene Fink

Quick reads, aimed at people creating quantitative studies, but the principles of research design and question construction apply to qualitative inquiry as well.

[Managing the Professional Service Firm](#)

David Maister

If you are running an R&D, design, or engineering group within a company, you would do well to think of yourselves as a professional services firm. If you are buying services from an outside firm, a good guide to what they are (or should be) doing, and where their interests lay.

[The Innovator's Dilemma](#)

[The Innovator's Solution](#)

Clay Christensen

Discusses the concept of "good enough" with respect to users' needs and desires, and the dynamics of competition this imposes on products.

[Wellsprings of Knowledge](#)

Dorothy Leonard

Discusses the concept of empathic design, and how innovative organizations build that capability.

[The Visual Display of Quantitative Information](#)

[Envisioning Information](#)

[Visual Explanations: Images and Quantities, Evidence and Narrative](#)

Edward Tufte

The definitive works on using visuals to promote clear thought and clear communication.

[Flatland: A Romance of Many Dimensions](#)

Edwin A. Abbott

A fun read, and a good exercise in thinking about how a society's ability to perceive shapes its activities, its views on what is "normal," and its capacity for understanding other cultures.

COURSE ASSIGNMENTS

See grading criteria and schedule.

COURSE GRADING

Your performance will be evaluated as follows:

Component	Percentage
Class Participation	30 %
Practical Exercises	30 %
Team Project Presentation (Presented in session 10)	40 %

Class Participation: Your class participation (contribution to class discussions) will count for 30% of the class grade. To help in preparation, I have suggested "Guiding Questions" for every class in the detailed syllabus. These questions should serve merely as a starting point. At the beginning of the class, one or more class members will be asked to start the session by addressing a specific question, or by providing an overview of the reading / preparatory exercise. Anyone who has prepared the case or done the exercise should be able to handle such a leadoff assignment. After a few minutes of initial analysis and recommendations, we will open the discussion to the rest of the class. As a group we will try to build a complete analysis of the situation. You are expected to be an active participant throughout the entire class and to contribute to the quality of the discussion. Please note that the frequency (i.e., the quantity) of your

interventions in class is not a key criterion for effective class contribution. Some criteria used to evaluate class contribution are as follows:

- Is the participant an active listener, looking for opportunities to enrich the discussion?
- Are the points that are made relevant to the discussion? Are they linked to the comments of others?
- Do the comments show evidence of analysis of the case / reading / exercise?
- Is there willingness to test new ideas, or are all comments "safe"? (For example, repetition of case facts without analysis and conclusions.)
- Do comments clarify or build upon the important aspects of earlier comments and lead to a clearer understanding of the case / reading / exercise?

Practical Exercises: New product development is cross-functional, time-sensitive, and competitive. As a result, leading-edge techniques are constantly being generated and refined, and most practitioners learn by doing. We will use individual and team exercises as a teaching tool, and to provide information for the final team project. There is no "right" answer for these, but because some of the exercises require outside prep, and because the individual contributions form the basis of a collective learning experience, I will be evaluating your preparation, intensity of application, and thoughtfulness of execution.

Attendance: In a hands-on, project-based class, it is very difficult to make up for an absence, because so much of the learning comes from your engagement with the activity in the moment, and your reflection upon it afterwards. You are expected to attend every class, and you are responsible for the material covered in class whether or not you attend. However, I realize that despite your best efforts you might be forced to miss a class due to unforeseen emergencies. Makeup for the session will depend upon the material covered. Because of our reliance on practical exercises and team work, you must attend the section in which you are enrolled, and you should realize that an absence affects classmates as well.

Team Project: There will be a team deliverable due the last class session which will count for a total of 40% of the class grade. It is intended to take you through a new product development cycle, though it is compacted to be manageable within the constraints of a university course and your schedules. The general topic will be the same for all teams, and the practical exercises will serve to provide some baseline information to inform the development of this new product.

I will expect your team to:

- further supplement this with additional research,
- synthesize an understanding of the environment of use,
- uncover opportunities and imperatives for a new product / service in that space,
- prototype critical elements of the new product / service, and
- communicate to the rest of us the vision for the product, and why it is compelling for both user and business.

The deliverables are in two parts, graded separately:

- A 10 to 20 minute in-class presentation to the class, and
- A report on the project in whatever medium the team desires.
- Please remember that quality, not quantity, is the key to success in the project.

We may set aside some class meeting time for team project work, and I will be available for consultations and work sessions with the teams. I am happy to help guide and critique your work along the way.

ACADEMIC INTEGRITY

Integrity of scholarship is essential for an academic community. As members of the Rady School, we pledge ourselves to uphold the highest ethical standards. The University expects that both faculty and students will honor this principle and in so doing protect the validity of University intellectual work. For students, this means that all academic work will be done by the individual to whom it is assigned, without unauthorized aid of any kind.

The complete UCSD Policy on Integrity of Scholarship can be viewed at:
<https://academicintegrity.ucsd.edu/process/policy.html>

STUDENTS WITH DISABILITIES

A student who has a disability or special need and requires an accommodation in order to have equal access to the classroom must register with the Office for Students with Disabilities (OSD). The OSD will determine what accommodations may be made and provide the necessary documentation to present to the faculty member.

The student must present the OSD letter of certification and OSD accommodation recommendation to the appropriate faculty member in order to initiate the request for accommodation in classes, examinations, or other academic program activities. **No accommodations can be implemented retroactively.**

Please visit the [OSD website](#) for further information or contact the Office for Students with Disabilities at (858) 534-4382 or osd@ucsd.edu.

COURSE SCHEDULE

The course is arranged to generally follow the flow of activities in a typical new product development cycle, with some adjustment to allow for exercises that must be prepared ahead outside of class. It is “front-loaded,” to ensure that students have the tools needed for the final project. This gives us some flexibility in the final sessions to explore specific topics in greater detail. Depending upon feedback from the class, we can explore topics such as organizational issues for large / small / entrepreneurial NPD, process differences in software and biotech/pharma, etc. I actively solicit your input on this.

Session (date)	Topics	Reading Assignments
1) Jan 08	Introduction Making the Case: Models & Frameworks Overview of the NPD process Assignment of Food Observation and In-Context Interview	• U&E Ch 1-3 • https://youtu.be/flKcN2x50rw • https://youtu.be/ykumbLhta5U • https://youtu.be/P5lty34oaQ (these videos are also on Canvas)
Study Questions	• Have you ever been part of a product development effort? What was the team structure? What was the project structure? What did you think worked well? What do you think worked poorly? • What do you think of the project structure Ulrich and Eppinger lay out in Ch 2? Does it make sense? Do you see anything missing? • Is there a “most important” step in the development process? Why or why not? • What do you think of the product planning process described in Ch 3? Would you characterize it as thorough or random? Customer-led or company-led? • What do you think of the “Jobs to Do” approach described in the videos? (we viewed them in L2M – I recommend that you review)	
Prep Exercises	• Define the term “product.” What exactly does it encompass? • “Meh” - Just an Average Product Choose a product from your everyday life that you use regularly, but neither like nor dislike. Be prepared to speak for a maximum of two minutes on exactly what job it performs for you, and submit a writeup through Canvas. (max one paragraph, include a photograph of the “meh” product) Please have the materials and the actual product with you when you come to class, and be prepared to present them.	
Session (date)	Topics	Reading Assignments
2) Jan 15	Empathic design Overview of the NPD process Design-a-Food Exercise I	• Henry Heinz: Making Markets for Processed Foods, HBS 9-801-289 • “Spark Innovation Through Empathic Design,” Dorothy Leonard & Jeffrey Rayport, Harvard Business Review, Nov-Dec 1997, Product 97606 • “Naked Truth Meets Market Research: Perfecting a New Shower Head? Try Watching People Shower.” Washington Post Article, Sunday, February 24, 2002.

Study Questions	• If you were Henry Heinz in 1865, with half interest in a brickyard, and three or four acres of farmland producing horseradish and miscellaneous vegetables, how would you think about opportunities for the future? • If you knew you wanted to pursue your food business, how would you figure out what to make? • What is it that Heinz is really selling? • How did Heinz manage to create this thing that he is selling? • How well or poorly does it fit into his customers' lives? • Is this a story of brand creation or product development?	
Prep Exercises	Observation and In-Context Interview • Pair up in teams of two to plan and conduct an Observation and In-Context Interview with a third non-Rady person as the participant (I suggest not your significant other), capturing their relationship with food – acquisition, preparation, consumption, recovery. Be sure to capture both functional and emotional elements. • Begin by constructing an interview protocol for your team – mind-mapping is a good tool to help you surface your own understanding, suppositions, and blind spots – so that you can know what activity you want to observe (preparation and consumption), what you expect to see, what might be surprising, and what you just don't know anything about. • This builds upon the food photo exercise in MGT 412, extending your skills to become a true field researcher, which is foundational for anyone seeking to practice User-Centered Design. • Capture your findings however you think best, and bring them to class. You will pair up with another team to share and further synthesize your findings, and use them as the foundation of another exercise.	
Session (date)	Topics	Reading Assignments
3) Jan 22	Cognitive Design Milk Carton Exercise Design-a-Food Exercise II Assignment of Photo Diary Exercise Assignment of Image Board Exercise	Affordances and Signifier Videos I suggest that you watch these videos before the assigned reading: • Throw the Box Intro to affordances, conceptual models, signifiers • Affordances & Signifiers Car door handles • Missing Signifiers Car doors • Signifier with No Perceived Affordance Tesla door handle

		• Signifiers & Missing Signifiers Video player Psychopathology of Everyday Things, Ch. 1 DOET The Psychology of Everyday Actions, Ch. 2 DOET
Study Questions	The videos above are not a substitute for the readings in The Design of Everyday Things (DOET), but if viewed first, they will aid in the reading. Be sure that you understand the concepts introduced, and how they relate to each other. Notice that they form natural pairings: Affordance – Signifier Mental Model – System Image Gulf of Execution – Gulf of Evaluation And that within each pair, the two terms are both related and distinct from each other. Each pair relates to the others as well. How is this? Be prepared to discuss this in class, and use what you have learned from the videos and readings to inform your work in the exercise.	
Prep Exercises	• Working as a team, continue the exercise we began in class last week. • Design a food product for one of the users that you studied. • Use the Insights --> Qualities --> Features --> Sketch framework that we learned in class, taking care to describe and show the relationship between the qualities that this user would find desirable in the product, and the features that deliver on that quality. • Upload your product on Canvas, and be ready to present and discuss in class.	

Session (date)	Topics	Reading Assignments
4) Jan 29	Design Research Techniques Analyzing qualitative findings Articulating opportunities	• Sony AIBO: The World's First Entertainment Robot, HBS 9-502-010 • "Margaret Mead Meets Consumer Fieldwork," Jennifer McFarland, Harvard Management Update, August 2001, Product U0108C • U&E Ch 5
Study Questions	<u>Aibo</u> • Why has AIBO been such a hit? (or has it been?) • Who are its customers, and why are they buying it? Is there a difference between the US and Japan? • Does AIBO fit with the rest of Sony's products? • What drove Sony's definition of what AIBO is and does, what AIBO is not and does not do? • Would you have defined AIBO the same way? If not, how would you have defined it? • Where the heck are they going with this? <u>Photo Diary / Image Board</u> • Is this "ethnography?" How accurate and actionable do you think the information gained actually is? • How does this relate to traditional marketing research?	
Prep Exercises	<u>Photo Diary</u> Perform the exercise for the topic you selected or were assigned (below). Be sure your booklet is complete and that you have a printed copy, as we will exchange and analyze them in class. <i>Transportation</i> <i>Relaxation</i> <u>Image Boards</u> For the opposite topic, bring in a dozen or so images from whatever source (printed out, if you found them online), that somehow are meaningful to you on that topic (that is, if your photojournal is about your car, look for images on relaxation). We will construct and analyze image boards in class.	

	<u>Design-a-Food Iteration</u> - Consider whether, and if so how, the concepts introduced in DOET might influence what your team came up with in the Design-a-Food exercise from the previous class. For example, what affordances does your user want or need? What signifiers did you create to guide them to notice and use the affordances? - Sketch your revised product, submit both the original and the iterated version on Canvas. - Be prepared to present in class, explaining your rationale; how and why has the product evolved?	
Session (date)	Topics	Reading Assignments
5) Feb 05	Articulating opportunities Service Design, Digital Design, and Prototyping	Knowledge in the Head and in the World , Ch. 3 DOET Knowing What to Do: Constraints, Discoverability, and Feedback , Ch. 4 DOET
Study Questions	- Consider the concepts discussed in this week's reading in The Design of Everyday Things – they represent the next level of understanding the application of Cognitive Design to real-world products. - As an exercise to test and solidify your understanding, think about a computer application you commonly use. Come up with a list of tasks you perform frequently, and that you perform infrequently. Run through those tasks, and notice how you know: a) what to do, b) how to do it, and c) whether it has been done as you wanted. - Take notes on your study exercise, preparing as if you needed to brief a co-worker about your observations. We will discuss your observations and insights in class.	
Prep Exercises	Parking: Experiential Learning → Service Design - As you interact with parking, on campus, at work, running errands, or just out and about over the next week, pay particular attention to things that work well and things that do not work well at all. - We will draw upon that understanding for an in-class exercise - As opposed to last quarter, you will not draw upon that understanding for “big picture” ideation. Instead, I will give you a design brief; we will learn and practice tools for designing services and digital interfaces Network / Ecosystem Analysis: Product I Love	

	• Bring to class a product you love – one that fits your life so well that you cannot imagine being without it. • Map out the network / ecosystem for the product. Be prepared to show a graphic representation of the network and discuss why you love the product. Try to link specific features of the product to its fulfillment of your needs / desires. • We have studied several different models and frameworks for relating a person's happiness with a product to the features it incorporates and how those features are actually implemented – these tools are all complimentary to each other, so think about how you might draw upon them to help explain this Product You Love • There are examples of this from previous classes in this folder. We will begin a practical exercise that will continue into the following class session, in the same teams that you will use for your final project • Be sure you have formed a team, and please email me the team roster before class. • The final project will be assigned as a general topic area, with some flexibility for the team to choose a specific focus. You will not be able to work on the same product or business idea that you have for another class or outside interest, so there is no need to make your team match any other class or outside interest. • The ideal team size is 3-5 people. Remember what we have learned about the value of diverse skillsets, perspectives, and backgrounds in improving the work of the team, as you consider your ideal composition.	
Session (date)	Topics	Reading Assignments
6) Feb 12	*** No Class This Week ***	•
Prep Exercises	park.ly exercise • Use this extra time to work on park.ly • Remember the importance of in-person collaboration for creative work, that all your teammates have our normal class time blocked out from other obligations, and that the classroom will be empty during that time as well...	
Session (date)	Topics	Reading Assignments
7) Feb 19	Product I Love continued Research and Development planning Assignment of Final Project	• U&E Ch 3, 5, 6 • Innovation at 3M Corp. (A), HBS 9-699-012 • Note on Lead User Research, HBS 9-699-014 • "Customers as Innovators: A New Way to Create Value," Stefan

		Tomke & Eric von Hippel, Harvard Business Review, April 2002, Product R0204F
Study Questions	A portion of these readings were previously assigned (U&E Ch 3), and I would like you to review them as the basis for our discussion in this session. The rest are new. We will be discussing how to actually plan and conduct research and development of the type we have studied, so think of questions you may have around "how does this work in real life?"	
Prep Exercises	We will begin a practical exercise that will continue into the following class session, in the same teams that you will use for your final project • Be sure you have formed a team, and please email me the team roster before class. • The project will be assigned as a general topic area, with some flexibility for the team to choose a specific focus. You will not be able to work on the same product / business idea that you have for another class or outside interest, so there is no need to make your team match any other class or outside interest. • The ideal team size is 3-5 people. Remember what we have learned about the value of diverse skillsets, perspectives, and backgrounds in improving the work of the team, as you consider your ideal composition.	
Session (date)	Topics	Reading Assignments
8) Feb 26	Prototyping – Physical and Digital Planning for User Research Introduction of Final Project	
In-class Exercise	Digital Prototyping – Be prepared to share and solicit feedback on the phone-based prototype you have built in your team. Physical Prototyping – We will conduct a physical prototyping exercise based upon your thoughts about a physical product / infrastructure to accompany your app.	
Prep Exercises	Revise your storyboards and wireframes based upon the feedback you received in class, as well as your own evaluation of your first iteration. Use the tools of Information Architecture and Content Inventory to inform your work. (You will not need to present these, but they will be vital to doing good work.) Construct phone-based click-through prototypes as discussed in class using the Prototyping on Paper (PoP) app.	

	Consider what physical products or infrastructure would be required or beneficial to deliver the experience you intend to provide for users. We will use this as an input to the physical prototyping exercise.	
Session (date)	Topics	Reading Assignments
9) Mar 05	Concept Generation Prototyping Constructive Critique Iteration and Refinement	• “Enlightened Experimentation: The New Imperative for Innovation,” Stefan Tomke, Harvard Business Review, Feb 2001, Product R0102D • “R&D Comes to Services: Bank of America's Pathbreaking Experiments,” Stefan Tomke, Harvard Business Review, April 2003, Product R0304E • U&E Ch 7, 8, 9 • U&E Ch 14
In-class Exercise	Prototyping – We will brainstorm, prototype, critique and refine a simple product in class. Details to be provided, builds off previous exercises.	
Prep Exercises	Internal Brainstorm / Fringe Users – Within your team, mind map on the attributes and functions of car or transportation, to uncover what you know about how people relate to their car and the things they carry or store in their car. Then make a map of the users you might want to observe to learn more about drivers' (or passengers') needs, working out from people very similar to drivers to people who may be different but who also face similar needs (though perhaps in more extreme ways). From this, select an area of focus that you think will be interesting for your Final Project. Observation and In-context Interview – Using the techniques discussed in Sessions 2, 3, and 4, conduct an in-context interview of at least one person who is not a Rady student, relative to how they use their car or other transportation, and as input to your Final Project. It must include a trip, a task or goal, and a digital interaction in support of the experience. You must accompany them through the entire arc. Pay particular attention to capturing the person's perceptions and thoughts in the digital interactions. <i>How will you do this, seeing the unseen?</i> Be prepared to share with the rest of the class, through images, video, and storytelling, what needs your team saw through these observations, and what insights this has given you.	

Session (date)	Topics	Reading Assignments
10) March 12	Final Presentations Visual Communication Flexible Topic	- Visual and Statistical Thinking: Displays of Evidence for Making Decisions, Edward Tufte, Graphics Press - Essay: The Cognitive Style of Powerpoint, Edward Tufte, Graphics Press
Study Questions	- What did Dr. Snow do right? What did NASA and Morton Thiokol do wrong? - Is this an issue of good science versus bad science, or good communication versus bad communication? (or is it something else entirely?) - Imagine yourself living in 1854 London. Louis Pasteur has not yet suggested to the scientific community that invisible animals called germs are responsible for infection and disease (this will happen in 1865). Would you have been convinced that removing a single water well pump-handle could stop this catastrophe? - Take note of the NASA charting faults on pages 47 and 48. Bring to class an example of one of these faults from another source. - How well does Tufte integrate text and graphics?	
Prep Exercise	Concept Generation, Prototyping, Testing, Iteration – Build upon your insights from the field work you conducted for Week 9, and the ideation you conducted in Session 9. Prototype your product, test with at least one user who is not a part of your team in the context of intended use, and refine your prototype based upon your testing. Final project presentation – Be prepared to present your team's final project to the rest of the class.	
Session (date)	Topics	Reading Assignments
TBD	Field Trip: D&K Engineering – or – UCSD MakerSpace	Tentative: Pending arrangements
Transportation	Carpool – We will meet at D&K Engineering's facilities in 4S Ranch / Rancho Bernardo. Our visit starts promptly at 10am. They are generously treating us to lunch. You should plan on a 45 minute drive to D&K from Rady in typical morning commute traffic. Our visit starts promptly at 10am.	
Prep Exercises	Design and Engineering Process / Prototyping / Manufacturing	

	We will be visiting https://www.dkengineering.com. Take some time beforehand to explore their website – it is quite informative, about process, capabilities, and portfolio, representative of the sort of partner you would consider working with in a real-world NPD effort. Think upon our discussion of prototyping, your reading on the topic, and the larger design and development process that we have been learning. What do you still want to understand better? What do you want to see in action? What do you want to know more about? Come to the field trip with an active curiosity, primed by reflection on the questions above, and informed by your background research on the company.
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